



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma White Belt (CLSSWB) Training

Course Code: C142

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 July 2026	Initial release — CLSSWB 1-day lesson plan for the Lean Six Sigma awareness course.	Dr. Alfred Ang
2	19 July 2026	Rebuilt from the single-source content module: DMAIC roadmap end to end within one 8-hour training day, exactly one hands-on lab per DMAIC phase (5 labs) on a single continuous scenario, content simplified to White Belt awareness depth.	Dr. Alfred Ang
3	21 July 2026	Released under course code C142 as a commercial one-day hands-on course. Schedule restructured so every DMAIC lab gains additional hands-on time and the day closes with a full course recap and Q&A.	Dr. Alfred Ang
4	21 July 2026	Slide deck expanded to full teaching depth: per-concept teaching slides for Foundations and every DMAIC phase (worked examples, diagrams and phase recaps), matching the depth of the original White Belt trainer deck.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Learning Reinforcement.....	4
Course Schedule.....	4
Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, hands-on.....	4
Lab Reference (aligned to course topics).....	5

Course Information

Course Title	Certified Lean Six Sigma White Belt (CLSSWB) Training
Course Reference	C142
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 8 training hours per day (8 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the BrewBean Cafe morning rush improvement scenario
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, hands-on

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction and ground rules
10:00–11:00	60 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma;

		the belt roles and where a White Belt contributes; the DMAIC roadmap
11:00–11:15	15 min	Tea break
11:15–12:00	45 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; problem statements; SMART goals; the project charter and scope
12:00–12:40	40 min	Hands-on: Lab 1: Define — Customer Requirements and the Problem Statement
12:40–13:40	60 min	Lunch break
13:40–14:25	45 min	DMAIC · MEASURE — process mapping; SIPOC; types of data; data collection plans; check sheets; the eight wastes (DOWNTIME); value-added analysis
14:25–15:05	40 min	Hands-on: Lab 2: Measure — Map the Process and Spot the Waste
15:05–15:50	45 min	DMAIC · ANALYZE — symptom vs root cause; 5 Whys; Fishbone (5M); reading a Pareto chart; common vs special cause variation
15:50–16:05	15 min	Tea break
16:05–16:45	40 min	Hands-on: Lab 3: Analyze — Find the Root Cause using the 5 Whys and Fishbone tools
16:45–17:10	25 min	DMAIC · IMPROVE — generating solutions; impact/effort screening; 5S; mistake proofing (Poka-Yoke); standard work; piloting
17:10–17:35	25 min	Hands-on: Lab 4: Improve — Choose and Pilot a Countermeasure
17:35–17:55	20 min	DMAIC · CONTROL — the control plan; visual management; SOPs; team huddles; handover
17:55–18:15	20 min	Hands-on: Lab 5: Control — Hold the Gain and Hand Over
18:15–18:30	15 min	Course recap and Q&A

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Lean Six Sigma Foundations	20%	
Topic 02: Define — Understand the Problem	20%	Lab 1

Topic 03: Measure — See What Is Really Happening	20%	Lab 2
Topic 04: Analyze — Find the Cause	20%	Lab 3
Topic 05: Improve — Fix the Cause	10%	Lab 4
Topic 06: Control — Hold the Gain	10%	Lab 5